

Problem - Solution Fit - ShopEZ Stock Trading Platform

Parameter	Details
Date	18 June 2026
Team ID	SE-09
Project Name	ShopEZ Stock Trading Platform
Maximum Marks	2 Marks

Problem - Solution Fit Template

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose

- Solve complex virtual trading sandboxes in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points by finding the right problem-behavior fit and building trust by solving frequent annoyances (e.g. slow updates, manual math).
- Understand the existing simulation situation in order to improve it for your target group.

Canvas Details

Target Groups

- **Traders (Users):** Individuals seeking to learn stock trading, test strategies, and view live market dashboards without risking real capital.
- **System Administrators:** Platform operations team managing listings updates, auditing transactions, and securing user registries.

Existing Behavior & Triggers

- **Behavior:** Traders maintain manual spreadsheet sheets to calculate virtual investments, view stock quotes on delayed public trackers, and use complex interfaces. Admins use direct database scripts to add tickers and manage configurations.
- **Triggers:** Stock market open, sudden stock price movements, need to test investment strategies before live broker deployment, and new user onboardings.

Problems & Pain Points

- **Trader Pain:** Outdated ticker prices, complicated charts, high risk of financial loss on real exchanges, and difficult calculations for average cost averages.
- **Admin Pain:** Manual ticker updates, lack of audit dashboards, and vulnerability to transaction database exploits.

Proposed Solutions

- **For Traders:** Real-time simulated stock price feed dashboard, a simple one-click trade form, and automated average price and profit/loss calculations.
- **For Admins:** Sleek administration portal with summaries (total users, active stocks, transaction log) and forms to add/remove tickers.

Channels & Touchpoints

- Responsive Web Application (Vite + React client, Express API server, MongoDB database).
- Dashboard interfaces optimized for mobile, tablet, and desktop browser screens.

Trust & Security

- **Admin Approval:** Stock listings are vetted and updated dynamically only through approved admin forms.
- **Secure Login:** Encrypted user passwords (bcryptjs) and secure stateless JWT sessions.
- **Portfolio Isolation:** Portfolio logs and transaction history details are private and restricted to the account owner.